

A concept class $\mathcal{C}$ is said to be PAC learnable if there exists a learning algorithm $\mathcal{A}$ and a polynomial function $\text{poly}(1/\epsilon, 1/\delta, \dots)$ such that for any $\epsilon > 0$, $\delta > 0$, any distributions $\mathcal{D}$ on $\mathcal{X}$ and any target concept $c \in \mathcal{C}$, the following holds for any sample size $m \geq \text{poly}(1/\epsilon, 1/\delta, \dots)$, $h_S = \mathcal{A}(S)$ then

$$\mathbb{P}_{S \sim \mathcal{D}^m}[L_{\mathcal{D},c}(h_S) \leq \epsilon] \geq 1 - \delta$$

That is, the hypothesis returned by the learning algorithm after observing a polynomial number of examples is approximately correct (within error ϵ) with high probability (at least $1 - \delta$).

0.0.1 See also

0.0.2 References